Part V

Post-processing operations

The unit operations described in preceding chapters are used to prepare foods for processing (Part II) or process them to extend their shelf life and/or alter their sensory characteristics (Parts II, III and IV). In this Part, the unit operations may be integral to previous operations (for example coating chilled or frozen foods, confectionery, etc. (Chapter 23)) or packaging foods to extend their shelf life and assist in promotion (Chapters 24 and 25). Other ancillary operations, including materials handling within a factory or warehouse, storage and distribution technologies and waste management and disposal (Chapter 26) are all critical to the success of commercial food processing.

23

Coating or enrobing

Coatings of batter or breadcrumbs are applied to fish, meats or vegetables, chocolate or compound coatings are applied to biscuits, cakes, confectionery and coatings of salt, sugar, flavourings or colourants are also applied to snackfoods, baked goods and confectionery. In each case, the aim is to improve the appearance and eating quality of foods, and to increase their variety. In some cases coatings also provide a barrier to the movement of moisture and gases, or protect the food against mechanical damage. Coatings are also applied to foods:

- to improve appearance
- to modify the texture
- to enhance flavours
- to improve convenience
- to increase variety and add value to basic products.

Coating operations have a minimal effect on the nutritional quality of foods, except in terms of the ingredients added to food in the coatings.

Methods of coating particles of food to encapsulate flavours or other ingredients are described by Dewettinck and Huyghebaert (1999).

There are three main methods of coating foods. The selection of an appropriate method depends on the type of coating material to be used and the intended effect of coating. The main methods are:

1. enrobing with chocolate, compound coatings, glazes or batters
2. dusting with spices, breadcrumbs, flour, sugar, flavourings, colourings, salt, etc.
3. pan coating with sugar or sugarless coatings.

23.1 Coating materials

There are two main types of coating material that are used to enrobe foods:

1. batters, breadcrumbs and powders for savoury foods and
2. chocolate, sugar or compound coatings for sweet foods such as confectionery, ice cream and baked goods.

23.1.1 Batters, powders and breadcrumbs

Batters are a suspension of flour in water to which various amounts of sugar, salt, thickening, flavourings and colourings are added to achieve the required characteristics. They are applied for example to fish, poultry and potato products. A single layer of viscous batter (termed 'Tempura') is used for products that are not subsequently breaded. A thinner, adhesive batter is applied to products prior to coating with crumb. Both are applied either by passing the product through batter using submerged mesh conveyors or by passing food pieces through one or more 'curtains' of batter in similar equipment to enrobers. Both types of equipment are fitted with air knives to remove excess batter and to control the thickness of the coating.

Typical examples of powder and crumb coatings are dry mixtures of spices, salt and flavourings applied to savoury foods, or sugar powder on sweet products, biscuits or cakes. Many of the flavour coatings, particularly barbecue dusts, are hygroscopic and require careful storage and handling in the dusting machine to prevent agglomeration and consequent depositing of large granules of dust onto the product.

Different types of crumb are available for breading fish, meat or vegetables, including wholemeal wheat or oats, sesame, and combinations of wheat, barley and rye. Maize crumb may be mixed with potato flakes to give a two-tone effect when applied to vegetables. Each are baked into a crumb with a known range of particle sizes and are flavoured or coloured if required. All are fragile and require delicate handling. Recent product developments include a fresh breadcrumb, which has a softer eating texture and enhanced colour definition, and coatings based on Asian or Oriental foods. One such example is satay, where a specially formulated coating is sprinkled onto pieces of meat or poultry and then grilled to produce a hot, spicy, peanut flavoured sauce (Marriott, 1989).

23.1.2 Chocolate and compound coatings

There are two main types of sweet coatings: chocolate and compound coatings. Chocolate manufacture is described in detail by Beckett (1994). Cocoa butter is a polymorphic fat (it crystallises in different forms or 'polymorphs'). Some forms are unstable and change from one to another if the temperature changes, resulting in a white 'bloom' on the surface of the product. There are four polymorphs:

1. γ form, melting at 17ºC
2. α form, melting at 21–24ºC (very unstable, formed by rapid cooling of liquid fat)
3. β^1 form, melting at 27–29ºC (from γ form. Unstable and most likely to be present, will slowly change back to γ form)
4. β form, melting at 34–35ºC (the stable form, correct tempering should maximise this form).

It is therefore necessary to temper and cool chocolate under controlled conditions to ensure that only the stable form of cocoa butter is present. Dark chocolate is tempered by first melting it at 46–49ºC and then cooling with constant stirring to 28–29ºC to form nuclei for growth of predominantly β crystals, although β^1 forms are also still present. It is then reheated to 31–32ºC to melt unstable β^1 crystals and it is held at that temperature during production (Minifie, 1988). Subsequent cooling is controlled so that it is not too rapid to produce only stable fat crystals and avoid the development of *bloom*. Milk chocolate is tempered at temperatures about 2ºC lower than those above. If higher ratios of cocoa butter are used to make chocolate thinner, there is a greater risk of bloom formation and small amounts of lecithin (up to 0.5%) are added to reduce this risk. The

use of lecithin and other additives during enrobing is described in detail by Lees and Jackson (1973) and Talbot (1994) gives further details of chocolate tempering.

Tempering machines are either tubular or plate heat exchangers (see Chapters 11 and 13) that have accurate temperature control of the heating water. A portable 'temper meter' can be used to routinely test the state of temper of chocolate on a production line by plotting a cooling curve and comparing it to a standard for the particular type of chocolate being tested (Fig. 23.1). The first part of the curve indicates removal of sensible heat and this is followed by a change in the slope (point of inflexion) as latent heat of crystallisation is released. Differences in the angle of the initial slope and the point of inflexion can be used to assess the degree of temper.

Owing to the relatively high price of cocoa butter, a number of fats have been developed which are termed 'cocoa butter equivalents' (or CBEs) and have similar properties to cocoa butter. They are permitted at levels of up to 5% in many countries. Compound coatings are made from other fats and cannot legally be used in chocolate. These do not require tempering as the fats are not polymorphic. The principal ingredients in a compound coating are fat, sugar, corn syrup, flavourings, fat-soluble colourings and emulsifiers, which are mixed in different formulations to achieve the desired properties. Corn syrup and starch are used to reduce the sweetness and cost of coatings. The particle size of the starch has an important effect on the texture and is closely controlled. The thickness of a coating is controlled by the fat content (more fat produces a lower viscosity), and the type and amount of emulsifier (Appendix C). The ratios of sugar, starch and fat are carefully controlled to achieve the required flow characteristics for application of the coating and the desired mouthfeel and taste in the final product. An

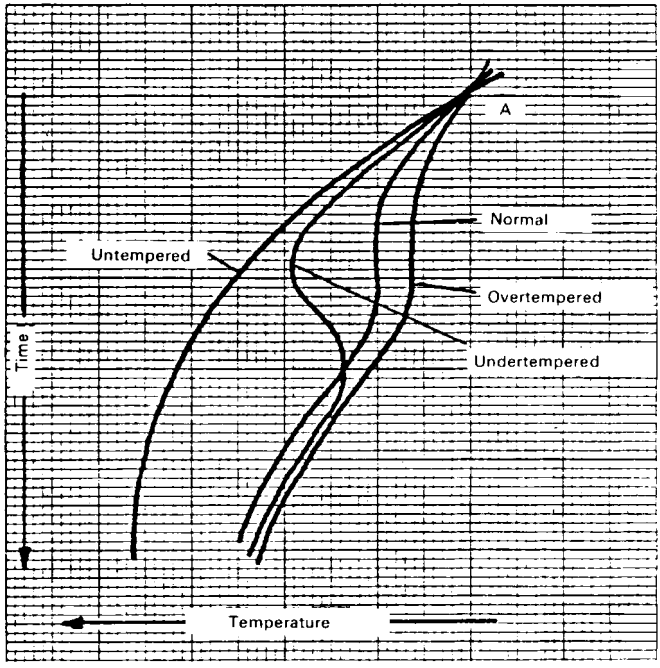


Fig. 23.1 Temper graph.
(Courtesy of Baker Perkins Ltd.)

example of the use of compound coatings is on cakes, such as Swiss roll, where the coating is more flexible than chocolate and will not chip off.

23.2 Enrobers

There are two types of enrober: in the 'submerger' type, food passes through batter on a stainless steel wire conveyor, held below the surface by a second mesh belt. In the second type, foods pass beneath a single or double curtain of hot liquid coating (Fig. 23.2). The coating is applied:

- by passing it through a slit in the base of a reservoir tank
- over the edge of the tank (spillway enrobers)
- by coating rollers.

A pan beneath the conveyor collects the excess coating and a pump recirculates it through a heater, back to the enrobing curtain. Excess coating is removed by air knives, shakers, licking rolls and 'anti-tailer' rollers to give a clean edge to the product. Discs, rollers or wires may be used to decorate the surface of the coating. A more detailed description of enrobers is given by Nelson (1994).

When enrobing products in chocolate, a separate first stage is termed 'pre-bottoming', in which the centres (e.g. peanuts) are passed on a wire belt through tempered chocolate.

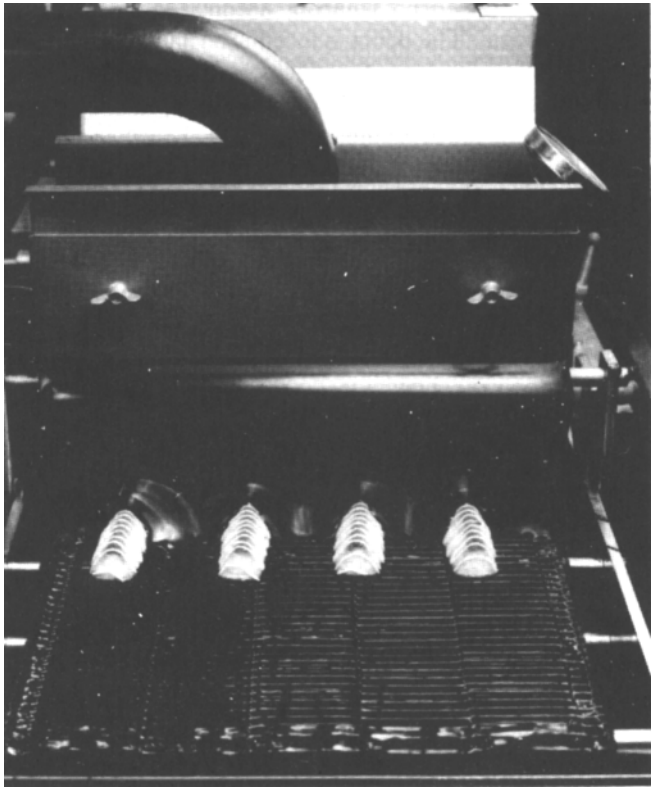


Fig. 23.2 Enrober.
(Courtesy of Sollich Ltd.)

The pre-bottomed centre then passes over a cooling plate to partially set the chocolate before passing through an enrober. This ensures that there is a sufficiently thick coating on the base of each centre to prevent it sinking through the chocolate. The centre is thus protected by the chocolate from moisture pickup or loss. The type and composition of centres can have a significant effect on the shelf life of enrobed confectionery. For example, centres should not be hotter than the enrobing temperature as this will cause a loss of temper. Nut centres should be sealed to prevent nut oil seeping into the casing and causing bloom formation (Lees and Jackson, 1973).

After enrobing, the coating is cooled by recirculated air in a cooling tunnel. Latent heat of crystallisation is removed from the product to prevent fat crystals from re-melting and coated foods are then held at 22ºC for 48 h to allow fat crystallisation to continue (Matz, 1972). Temperature zones are used to cool the product rapidly but not too rapidly to cause overcooling which in chocolate would produce surface bloom. The thickness of the coating is determined by:

- the temperature of both the food and the coating
- the viscosity of the coating
- the speed of the air in air blowers
- the rate of cooling.

23.3 Dusting or breading

Dusting or breading equipment consists of a hopper fitted with a mesh base, located over a conveyor. The mesh screen is changeable for different types of crumb, seasoning, flavourings, etc. In breading, foods that are coated with a thin batter (Section 23.1) pass on a stainless steel wire belt through a bed of breadcrumbs to coat the base and then through a curtain of crumb to coat the upper surface. Excess material is collected and conveyed back to the hopper by an auger or an elevator (Chapter 26).

Most coatings and dusts can be readily recirculated, but care is needed when handling crumbs to avoid damage and changes to the average particle size. Breaded foods are gently pressed between ‘tamping’ rollers to drive the crumb into the batter and to absorb batter into the material to create a strong bond. Excess crumb is removed by air knives and reused. After breading, the products are then frozen (Chapter 21) or fried and chilled (Chapters 17 and 19). Similar designs of applicator are used to coat confectionery and baked goods with sugar, flaked nuts or dried fruits.

In another design of coating equipment, a rotating stainless steel drum, slightly inclined from the horizontal, is fitted internally with angled flights or ribs to tumble the food gently and to coat all surfaces with seasoning. The angle and speed of rotation are adjusted to control the product throughput. Similar equipment is used for spraying products with oil or liquid flavourings. In other designs, flavours or salt are blown directly into the drum by compressed air. Fluidised beds and Torbed equipment (Chapter 15) are also used as flavour applicators.

23.4 Pan coating

Panning is the process of building up in a controlled way layers of sugar, sweetener or chocolate coating on cores of fondant, fruit, nuts, etc., using a revolving copper or

stainless steel pan. These products are characterised by a smooth, regular surface obtained by the polishing action in the pan. Panning is a slow process involving small batches, but with automatic operation one operator can monitor a bank of ten or more pans. Pan speeds vary according to the size of the centre: large nuts for example requiring speeds of 15 rpm and sugar grains ('hundreds and thousands') speeds of 30–35 rpm. Air at 35–65ºC is blown into the pan to give rapid drying of sugar layers, to remove dust and to remove frictional heat. During panning, the pieces are periodically removed and sieved to remove waste and break up any clumps. Additionally a small block of smooth wood in the pan helps to prevent clumping. There are three main types of pan-coated products, depending on the type of coating used.

23.4.1 Hard coatings

Centres are coated with a sweetener solution (termed 'wetting') which is added at a rate of 10–15% of the weight of the centres. This crystallises in successive layers and a hard coating is built up around the centres (termed 'engrossing'). If nuts are used, they should first be sealed with gum arabic/wheat flour mixture to prevent oil seepage during storage. Coatings are traditionally made from 60–65% sucrose or dextrose syrups. More recently sugarless coatings have been made from sorbitol syrup, described in detail by Le Bot (1993). Flavouring is added to each charge of wetting syrup and colouring is added in increasing concentration to the last five or six wettings. Cornflour may be added after each wetting to reduce sticking of the pieces but over-use results in agglomeration of the product. Separate glass-lined polishing pans are located away from the humid conditions in a pan room and are kept free of dust. Beeswax, paraffin oil or canauba wax are used to coat polishing pans and to shine hard pan products. Alternatively, they may be glazed using a mixture of shellac in isopropanol.

23.4.2 Soft coatings

Complex mixtures of liquid syrup and crystalline sugar are prepared as the centres for soft-coated confectionery such as jelly beans, 'Dolly Mixtures', etc. Because of the use of anti-crystallising agents in the liquid phase, the outer layers will only partially crystallise when added during pan-coating. Successive wettings of 60% glucose syrup are therefore followed by addition of fine castor sugar until the surfaces dry and produce an amorphous soft coating. The hardness of the coating is determined by the ratio of anti-crystallising agents in the syrup. The process should be stopped after two or three stages and the partially coated centres removed and allowed to dry for 2–3 hours. The final coating is dried using icing sugar and the products are then dried for two days at 20ºC in a dust-free room (Lees and Jackson, 1973). Sugarless soft coatings made from sorbitol, mannitol, maltitol and xylitol are described by Le Bot (1993).

23.4.3 Chocolate coating

This type of panning is similar to that used for hard coatings. Pans are rotated at around 20 rpm and held at 16ºC for plain chocolate and 14ºC for milk chocolate. Tempered chocolate may be poured or spray-coated onto confectionery centres with successive layers being built up and finished with a hard glaze, or polished with a 50% solution of gum arabic. A summary of the properties of this and other hydrocolloids used in confectionery is shown in Table 23.1.

Table 23.1 Some characteristics of hydrocolloids used in confectionery products

Characteristic/ source	Agar From red seaweed	Gelatin From animal tissues	Gum arabic From tree products	Pectin From apple pomace or citrus peel	Starch From maize
Usage levels in confectionery products	1–2	6–10	20–50	1–2	10–30
Solubilisation temperature (ºC)	90–95	50–60	20–25	70–85	70–85
Setting temperature (ºC)	35–40	30–35	20–35	75–85	20–35
Setting time (h)	12–16	12–16	24	1	12
Textural characteristics	short, tender	elastic, firm	very firm	short, tender, clean bite	soft to firm, chewy
Can be used in combination with	starch, gelatin	pectin, starch, gum arabic	starch, gelatin	starch, gelatin	gelatin, gum arabic, pectin

From Carr *et al*. (1995).

23.5 Acknowledgements

Grateful acknowledgement is made for information supplied by: Coat and Fry Ltd, Derby DE2 8JD, UK; Allen Machinery Systems, Stourbridge, West Midlands DY9 7NB, UK; Baker Perkins Ltd, Peterborough PE3 6TA, UK; Torftech Ltd, Mortimer, Reading, Berkshire RG7 3PG, UK; Interfood Ltd, Hemel Hempstead, Hertfordshire HP2 7DU, UK; Robinson's of Derby, Derby DE2 6JL, UK; Sollich GmbH and Co. KG, D-4902 Bad Salzuflen, West Germany; Sandvik Jahn, Huntingdon PE18 7EW, UK; Neilsen Ltd, c/o Robinson's of Derby Ltd, Derby DE2 6JL, UK; MOPA S.r.l., 1-44044 Cassana (FE), Italy; Spray Dynamics, Costa Mesa, CA, USA.

23.6 References

BECKETT, S.T. (1994) *Industrial Chocolate Manufacture and Use*, 2nd edn. Blackie Academic and Professional, London.

CARR, J.M., SUFFERLING, K. and POPPE, J. (1995) Hydrocolloids and their use in the confectionery industry. *Food Technology* (July), 41–44.

DEWETTINCK, K. and HUYGHEBAERT, A. (1999) Fluidized bed coating in food technology. *Trends in Food Science and Technology* **10**, 163–168.

LE BOT, Y. (1993) Stable sugarless coating. In: A. Turner (ed.) *Food Technology International Europe*. Sterling Publications International, London, pp. 67–70.

LEES, R. and JACKSON, E.B. (1973) *Sugar Confectionery and Chocolate Manufacture*. Leonard Hill, Glasgow, pp. 139–151 and 324–332.

MARRIOTT, D. (1989) Food coating processes. In: A. Turner (ed.) *Food Technology International Europe*. Sterling Publications International, London, pp. 67–71.

MATZ, S.A. (1972) *Bakery Technology and Engineering*. AVI, Westport, Connecticut, pp. 237–257.

MINIFIE, B.W. (1988) Methods of applying chocolate coating. In: A. Turner (ed.) *Food Technology International Europe*. Sterling Publications International, London, pp. 93–96.

NELSON, R.B. (1994) Enrobers, moulding equipment, coolers and panning. In: S.T. Beckett (ed.) *Industrial Chocolate Manufacture and Use*, 2nd edn. Blackie Academic and Professional, London, pp. 211–241.

TALBOT, G. (1994) Chocolate temper. In: S.T. Beckett (ed.) *Industrial chocolate manufacture and use*, 2nd edn. Blackie Academic and Professional, London, pp. 156–166.